

केस स्टडी/ CASE STUDY

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INTRODUCTION

- Title:** Fatal incident during transportation of HDPE wax-filled jumbo bag by forklift
- Location:** Road between the Wax Draining Station and Wax Storage Yard (approx. 140 mtrs), HDPE Unit of a Petrochemical Complex
- Loss/ Outcome:** Fatality of forklift operator and damage/toppling of forklift

BRIEF OF INCIDENT:

A fatal incident occurred during transportation of a wax-filled jumbo bag by forklift from the Wax Draining Station to the Wax Storage Yard in the HDPE Unit of a petrochemical Complex. The Wax Storage Yard was located approximately **140 m** from the Wax Draining Station. The filled jumbo bags, weighing approximately **500 kg each**, were routinely transported through the main road (11 m wide).



On the date of incident, the forklift operator, was engaged in shifting the filled wax bags in night shift. After successfully transporting the first bag of the batch, the operator proceeded with the second bag.

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When the forklift did not return within the normal travel time of approximately 4–5 minutes, the Production Junior Executive Assistant and contractor helper proceeded towards the Wax Storage Yard to ascertain his whereabouts. The forklift was found by them toppled inside an open trapezoidal dry drain approximately 1.0 m deep and 2.4 m wide, opposite the Wax Storage Yard on the other side of the lane in which the forklift was supposed to normally traverse. The operator was found unconscious beside the toppled forklift. He was then shifted to the First Aid Centre and next to nearby Civil Hospital, where he was declared brought dead.

The investigation identified several deficiencies relating to forklift load movement over an extended route, absence of roadside protection, inadequate traffic management, wax deposits on the roadway, operator fatigue/alertness, and the overall philosophy of repeated transportation of wax-filled bags by forklift.

OBSERVATIONS / LAPSES

1. The Wax Storage Yard was not envisaged at the original design stage of the HDPE Unit and was subsequently constructed outside the HDPE Unit boundary. The storage yard was located approximately 140 m from the Wax Draining Station.
2. Filled HDPE wax jumbo bags were being repeatedly transported by forklift from the Wax Draining Station to the Wax Storage Yard over the extended route. Approximately three batches per shift, involving around 12 jumbo bags, were transported.
3. Each one-way forklift trip took approximately 4–5 minutes,
4. The traversing road was observed to have considerable wax deposits on its surface, apparently resulting from spillage from the wax-filled jumbo bags. The roadway had an open trapezoidal storm-water channel approximately 1 m deep along its side, without adequate roadside protection/curb arrangement to prevent unintended vehicle movement into the drain.
5. No designated forklift travel route, road markings, warning signs or other adequate traffic management measures were evident along the main road despite regular forklift movement.
6. Although the work permit required compliance with the forklift SOP and marked route, neither the SOP nor the marked route was evident.
7. The affected forklift was owned by Organisation and had been procured in June 2021. However, the forklift was reportedly not registered with the concerned RTO as it was for internal use only. A Certificate of Inspection for the lifting equipment was available.
8. The Daily Running Service Sheet for forklifts did not include verification of the condition and functionality of the seat belt. During site inspection, the seat belt of the affected forklift was found to be non-functional. The seat belt of another forklift examined during the investigation was also found to be non-functional. This implies that IP was not wearing seat belt at the time of accident.
9. Two Toolbox Talk records, done approximately one and a half months apart, relating to wax draining and wax bag shifting operations were available; however, the involved forklift operator was not a participant as per the attendance records.
10. The operator was deployed for nine (09) consecutive night-shift forklift operations.
11. The existing arrangement involved transportation of semi-liquid wax slurry in jumbo bags directly on the forks of the forklift without wooden pallets. The practice was reportedly confirmed as acceptable by the jumbo-bag manufacturer.
12. The existing philosophy of transporting wax-filled jumbo bags by forklift, including the extended transportation route and storage arrangement, had not been adequately reviewed considering the

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load characteristics, frequency of transportation, road conditions, wax spillage and associated vehicle stability risks.

CONCLUSION / ROOT CAUSE

The likely cause is as follows:

1. Reduced operator alertness and situational awareness associated with prolonged consecutive night-shift deployment, affecting judgement and control of the forklift resulting in the forklift veering off towards the drain channel.
However, the following cannot be ruled off.
2. Probable turning of the forklift towards left at high-speed resulting in the hanging load sway towards the right resulting in loss of vehicle stability and its swerving towards the drain channel on the right.
3. Reduced tyre–road traction due to wax deposits/spillage on the roadway, adversely affecting the stability and manoeuvrability of the forklift.

RECOMMENDATIONS

1. The material-handling arrangement for transportation of wax-filled jumbo bags over extended routes shall be reviewed and suitable equipment specifically suited for repetitive horizontal transportation shall be considered, taking into account the associated traffic, collision and load-stability risks.
2. Pre-use inspection of forklifts shall be carried out before every shift, covering seat belts and other critical safety devices, and forklifts with defective safety devices shall not be deployed until the deficiencies are rectified.
3. Shift scheduling and rotation shall be reviewed to minimise fatigue arising from prolonged consecutive night-shift deployment, particularly for personnel engaged in safety-critical activities. Working hours, overtime and weekly rest shall be monitored in accordance with applicable statutory requirements.
4. Activity-specific Toolbox Talks shall be conducted prior to deployment of personnel on such operations, and participation of the concerned personnel shall be ensured through documented attendance records.
5. Appropriate roadside protection, such as curb stones, crash barriers or equivalent vehicle restraint systems, shall be provided along identified forklift travel routes adjacent to storm water drains or other roadside hazards to prevent unintended vehicular movement.
6. Adequate CCTV coverage shall be provided along designated forklift travel routes and critical material-handling areas, with suitable camera positioning and recording facilities to facilitate effective monitoring of forklift movement, traffic conditions and unsafe practices.
